

IT1223 (P) - Database Management Systems (P)

Department of Physical Science

University of Vavuniya

Revision Work Sheet – 05

QUESTION 1

1. Create the table Loan_Accounts and insert tuples in it.

```
CREATE TABLE Loan_Accounts (
  AccNo INT PRIMARY KEY,
  Cust_Name VARCHAR(50),
  Loan_Amount INT,
  Instalments INT,
  Int_Rate DECIMAL(5,2),
  Start_Date DATE,
  Interest DECIMAL(5,2)
);
```

```
INSERT INTO Loan_Accounts (AccNo, Cust_Name, Loan_Amount, Instalments, Int_Rate, Start_Date, Interest)
VALUES
(1, 'R.K. Gupta', 300000, 36, 12.00, '2009-07-19', 12.00),
(2, 'S.P. Sharma', 500000, 48, 10.00, '2008-03-22', 10.00),
(3, 'K.P. Jain', 400000, 56, NULL, '2009-03-08', NULL),
(4, 'M.P. Yadav', 800000, 60, 10.00, '2008-12-06', 10.00),
(5, 'S.P. Sinha', 600000, 60, 12.50, '2010-01-01', 12.50),
(6, 'P. Sharma', 700000, 60, 12.50, '2008-06-05', 12.50),
(7, 'K.S. Dhall', 500000, 48, NULL, '2008-03-05', NULL);
```

OUTPUT:

```
MariaDB [(none)]> CREATE DATABASE LOANS;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> USE LOANS;
Database changed
MariaDB [LOANS]> CREATE TABLE Loan_Accounts (
->   AccNo INT PRIMARY KEY,
->   Cust_Name VARCHAR(50),
->   Loan_Amount INT,
->   Instalments INT,
->   Int_Rate DECIMAL(5,2),
->   Start_Date DATE,
->   Interest DECIMAL(5,2)
-> );
Query OK, 0 rows affected (0.018 sec)

MariaDB [LOANS]> INSERT INTO Loan_Accounts (AccNo, Cust_Name, Loan_Amount, Instalments, Int_Rate, Start_Date, Interest) VALUES
-> (1, 'R.K. Gupta', 300000, 36, 12.00, '2009-07-19', 12.00),
-> (2, 'S.P. Sharma', 500000, 48, 10.00, '2008-03-22', 10.00),
-> (3, 'K.P. Jain', 400000, 56, NULL, '2009-03-08', NULL),
-> (4, 'M.P. Yadav', 800000, 60, 10.00, '2008-12-06', 10.00),
-> (5, 'S.P. Sinha', 600000, 60, 12.50, '2010-01-01', 12.50),
-> (6, 'P. Sharma', 700000, 60, 12.50, '2008-06-05', 12.50),
-> (7, 'K.S. Dhall', 500000, 48, NULL, '2008-03-05', NULL);
Query OK, 7 rows affected (0.015 sec)
Records: 7 Duplicates: 0 Warnings: 0
```

2. Display the details of all the loans with less than 40 instalments.

```
SELECT * FROM Loan_Accounts WHERE Instalments < 40;
```

OUTPUT:

```
MariaDB [loans]> SELECT * FROM Loan_Accounts WHERE Instalments < 40;
+-----+-----+-----+-----+-----+-----+
| AccNo | Cust_Name | Loan_Amount | Instalments | Int_Rate | Start_Date | Interest |
+-----+-----+-----+-----+-----+-----+
|      1 | R.K. Gupta |      300000 |          36 |      12.00 | 2009-07-19 |      12.00 |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.004 sec)
```

3. Display the AccNo and Loan_Amount of all the loans started before 01-04-2009

```
SELECT AccNo, Loan_Amount FROM Loan_Accounts WHERE Start_Date < '2009-04-01';
```

OUTPUT:

```
MariaDB [loans]> SELECT AccNo, Loan_Amount FROM Loan_Accounts WHERE Start_Date < '2009-04-01';
+-----+-----+
| AccNo | Loan_Amount |
+-----+-----+
|      2 |      500000 |
|      3 |      400000 |
|      4 |      800000 |
|      6 |      700000 |
|      7 |      500000 |
+-----+-----+
5 rows in set (0.001 sec)
```

4. Display the Int_Rate of all the loans started after 01-04-2009.

```
SELECT Int_Rate FROM Loan_Accounts WHERE Start_Date > '2009-04-01';
```

OUTPUT:

```
MariaDB [loans]> SELECT Int_Rate FROM Loan_Accounts WHERE Start_Date > '2009-04-01';
+-----+
| Int_Rate |
+-----+
|      12.00 |
|      12.50 |
+-----+
2 rows in set (0.001 sec)
```

5. Display the details of all the loans whose rate of interest is NULL.

```
SELECT * FROM Loan_Accounts WHERE Int_Rate IS NULL;
```

OUTPUT:

```
MariaDB [loans]> SELECT * FROM Loan_Accounts WHERE Int_Rate IS NULL;
```

AccNo	Cust_Name	Loan_Amount	Instalments	Int_Rate	Start_Date	Interest
3	K.P. Jain	400000	56	NULL	2009-03-08	NULL
7	K.S. Dhall	500000	48	NULL	2008-03-05	NULL

```
2 rows in set (0.001 sec)
```

6. Display the number of instalments of various loans from the table Loan_Accounts. An instalment should appear only once.

```
SELECT DISTINCT Instalments FROM Loan_Accounts;
```

OUTPUT:

```
MariaDB [loans]> SELECT DISTINCT Instalments FROM Loan_Accounts;
```

Instalments
36
48
56
60

```
4 rows in set (0.001 sec)
```

7. Display the Cust_Name and Loan_Amount for all the loans for which the loan amount is less than 500000 or int_rate is more than 12.

```
SELECT Cust_Name, Loan_Amount FROM Loan_Accounts WHERE Loan_Amount < 500000 OR Int_Rate > 12;
```

OUTPUT:

```
MariaDB [loans]> SELECT Cust_Name, Loan_Amount FROM Loan_Accounts WHERE Loan_Amount < 500000 OR Int_Rate > 12;
```

Cust_Name	Loan_Amount
R.K. Gupta	300000
K.P. Jain	400000
S.P. Sinha	600000
P. Sharma	700000

```
4 rows in set (0.001 sec)
```

8. Display the details of all the loans whose rate of interest is in the range 11% to 12%.

```
SELECT * FROM Loan_Accounts WHERE Int_Rate BETWEEN 11 AND 12;
```

OUTPUT:

```
MariaDB [loans]> SELECT * FROM Loan_Accounts WHERE Int_Rate BETWEEN 11 AND 12;
+-----+-----+-----+-----+-----+-----+-----+
| AccNo | Cust_Name | Loan_Amount | Instalments | Int_Rate | Start_Date | Interest |
+-----+-----+-----+-----+-----+-----+-----+
|      1 | R.K. Gupta |      300000 |          36 |      12.00 | 2009-07-19 |      12.00 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.001 sec)
```

9. Display the Cust_Name and Loan_Amount for all the loans for which the number of instalments are 24, 36, or 48. (Using IN operator).

```
SELECT Cust_Name, Loan_Amount FROM Loan_Accounts WHERE Instalments IN (24, 36, 48);
```

OUTPUT:

```
MariaDB [loans]> SELECT Cust_Name, Loan_Amount FROM Loan_Accounts WHERE Instalments IN (24, 36, 48);
+-----+-----+
| Cust_Name | Loan_Amount |
+-----+-----+
| R.K. Gupta |      300000 |
| S.P. Sharma |      500000 |
| K.S. Dhall |      500000 |
+-----+-----+
3 rows in set (0.001 sec)
```

10. Display the details of all the loans whose Loan_Amount is in the range 400000 to 500000. (Using BETWEEN operator)

```
SELECT * FROM Loan_Accounts WHERE Loan_Amount BETWEEN 400000 AND 500000;
```

OUTPUT:

```
MariaDB [loans]> SELECT * FROM Loan_Accounts WHERE Loan_Amount BETWEEN 400000 AND 500000;
+-----+-----+-----+-----+-----+-----+-----+
| AccNo | Cust_Name | Loan_Amount | Instalments | Int_Rate | Start_Date | Interest |
+-----+-----+-----+-----+-----+-----+-----+
|      2 | S.P. Sharma |      500000 |          48 |      10.00 | 2008-03-22 |      10.00 |
|      3 | K.P. Jain |      400000 |          56 |      NULL | 2009-03-08 |      NULL |
|      7 | K.S. Dhall |      500000 |          48 |      NULL | 2008-03-05 |      NULL |
+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.001 sec)
```

11. Display the AccNo, Cust_Name, and Loan_Amount for all the loans for which the Cust_Name ends with 'Sharma'.

```
SELECT AccNo, Cust_Name, Loan_Amount FROM Loan_Accounts WHERE Cust_Name LIKE '%Sharma';
```

OUTPUT:

```
MariaDB [loans]> SELECT AccNo, Cust_Name, Loan_Amount FROM Loan_Accounts WHERE Cust_Name LIKE '%Sharma';
```

AccNo	Cust_Name	Loan_Amount
2	S.P. Sharma	500000
6	P. Sharma	700000

```
2 rows in set (0.001 sec)
```

12. Display the AccNo, Cust_Name, and Loan_Amount for all the loans for which the Cust_Name contains 'a'.

```
SELECT AccNo, Cust_Name, Loan_Amount FROM Loan_Accounts WHERE Cust_Name LIKE '%a%';
```

OUTPUT:

```
MariaDB [loans]> SELECT AccNo, Cust_Name, Loan_Amount FROM Loan_Accounts WHERE Cust_Name LIKE '%a%';
```

AccNo	Cust_Name	Loan_Amount
1	R.K. Gupta	300000
2	S.P. Sharma	500000
3	K.P. Jain	400000
4	M.P. Yadav	800000
5	S.P. Sinha	600000
6	P. Sharma	700000
7	K.S. Dhall	500000

```
7 rows in set (0.001 sec)
```

13. Display the AccNo, Cust_Name, and Loan_Amount for all the loans for which the Cust_Name does not contain 'p'.

```
SELECT AccNo, Cust_Name, Loan_Amount FROM Loan_Accounts WHERE Cust_Name NOT LIKE '%P%';
```

OUTPUT:

```
MariaDB [loans]> SELECT AccNo, Cust_Name, Loan_Amount FROM Loan_Accounts WHERE Cust_Name NOT LIKE '%P%';
```

AccNo	Cust_Name	Loan_Amount
7	K.S. Dhall	500000

```
1 row in set (0.001 sec)
```

14. Display the AccNo, Cust_Name, and Loan_Amount for all the loans for which the Cust_Name contains 'a' as the second last character

```
SELECT AccNo, Cust_Name, Loan_Amount FROM Loan_Accounts WHERE Cust_Name LIKE '%a_';
```

OUTPUT:

```
MariaDB [loans]> SELECT AccNo, Cust_Name, Loan_Amount FROM Loan_Accounts WHERE Cust_Name LIKE '%a_';
```

AccNo	Cust_Name	Loan_Amount
4	M.P. Yadav	800000

```
1 row in set (0.001 sec)
```

15. Display the details of all the loans in the ascending order of their Loan_Amount and within Loan_Amount in the descending order of their Start_Date.

```
SELECT * FROM Loan_Accounts ORDER BY Loan_Amount ASC, Start_Date DESC;
```

OUTPUT:

```
MariaDB [loans]> SELECT * FROM Loan_Accounts ORDER BY Loan_Amount ASC, Start_Date DESC;
```

AccNo	Cust_Name	Loan_Amount	Instalments	Int_Rate	Start_Date	Interest
1	R.K. Gupta	300000	36	12.00	2009-07-19	12.00
3	K.P. Jain	400000	56	NULL	2009-03-08	NULL
2	S.P. Sharma	500000	48	10.00	2008-03-22	10.00
7	K.S. Dhall	500000	48	NULL	2008-03-05	NULL
5	S.P. Sinha	600000	60	12.50	2010-01-01	12.50
6	P. Sharma	700000	60	12.50	2008-06-05	12.50
4	M.P. Yadav	800000	60	10.00	2008-12-06	10.00

```
7 rows in set (0.001 sec)
```

16. Update the interest rate 11.50% for all the loans for which interest rate is NULL.

```
UPDATE Loan_Accounts SET Int_Rate = 11.50 WHERE Int_Rate IS NULL;
```

OUTPUT:

```
MariaDB [loans]> UPDATE Loan_Accounts SET Int_Rate = 11.50 WHERE Int_Rate IS NULL;
Query OK, 2 rows affected (0.009 sec)
Rows matched: 2 Changed: 2 Warnings: 0
```

17. Increase the interest rate by 0.5% for all the loans for which the loan amount is more than 400000.

```
UPDATE Loan_Accounts SET Int_Rate = Int_Rate + 0.5 WHERE Loan_Amount > 400000;
```

OUTPUT:

```
MariaDB [loans]> UPDATE Loan_Accounts SET Int_Rate = Int_Rate + 0.5 WHERE Loan_Amount > 400000;
Query OK, 5 rows affected (0.006 sec)
Rows matched: 5 Changed: 5 Warnings: 0
```

18. Add another column Category of type CHAR(1) in the Loan table.

```
ALTER TABLE Loan_Accounts ADD COLUMN Category CHAR(1);
```

OUTPUT:

```
MariaDB [LOANS]> ALTER TABLE Loan_Accounts ADD COLUMN Category CHAR(1);
Query OK, 0 rows affected (0.012 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [LOANS]> SELECT * FROM Loan_Accounts;
```

AccNo	Cust_Name	Loan_Amount	Instalments	Int_Rate	Start_Date	Interest	Category
1	R.K. Gupta	300000	36	12.00	2009-07-19	12.00	NULL
2	S.P. Sharma	500000	48	10.50	2008-03-22	10.00	NULL
3	K.P. Jain	400000	56	11.50	2009-03-08	NULL	NULL
4	M.P. Yadav	800000	60	10.50	2008-12-06	10.00	NULL
5	S.P. Sinha	600000	60	13.00	2010-01-01	12.50	NULL
6	P. Sharma	700000	60	13.00	2008-06-05	12.50	NULL
7	K.S. Dhall	500000	48	12.00	2008-03-05	NULL	NULL

7 rows in set (0.004 sec)

19. Delete the records of all the loans of 'K.P. Jain'.

```
DELETE FROM Loan_Accounts WHERE Cust_Name = 'K.P. Jain';
```

OUTPUT:

```
MariaDB [LOANS]> DELETE FROM Loan_Accounts WHERE Cust_Name = 'K.P. Jain';
Query OK, 1 row affected (0.007 sec)
```

```
MariaDB [LOANS]> SELECT * FROM Loan_Accounts;
```

AccNo	Cust_Name	Loan_Amount	Instalments	Int_Rate	Start_Date	Interest	Category
1	R.K. Gupta	300000	36	12.00	2009-07-19	12.00	NULL
2	S.P. Sharma	500000	48	10.50	2008-03-22	10.00	NULL
4	M.P. Yadav	800000	60	10.50	2008-12-06	10.00	NULL
5	S.P. Sinha	600000	60	13.00	2010-01-01	12.50	NULL
6	P. Sharma	700000	60	13.00	2008-06-05	12.50	NULL
7	K.S. Dhall	500000	48	12.00	2008-03-05	NULL	NULL

6 rows in set (0.001 sec)

QUESTION 2

Consider the Empl table and write SQL command to get the following:

```
MariaDB [LOANS]> CREATE TABLE Empl (
->   empno INT PRIMARY KEY,
->   ename VARCHAR(50),
->   job VARCHAR(50),
->   mgr INT,
->   hiredate DATE,
->   sal DECIMAL(10, 2),
->   comm DECIMAL(10, 2),
->   deptno INT
-> );
Query OK, 0 rows affected (0.021 sec)

MariaDB [LOANS]> INSERT INTO Empl (empno, ename, job, mgr, hiredate, sal, comm, deptno) VALUES
-> (8369, 'SMITH', 'CLERK', 8902, '1990-12-18', 800.00, NULL, 20),
-> (8499, 'ANYA', 'SALESMAN', 8698, '1991-02-20', 1600.00, 300.00, 30),
-> (8521, 'SETH', 'SALESMAN', 8698, '1991-02-22', 1250.00, 500.00, 20),
-> (8566, 'MAHADEVAN', 'MANAGER', 8839, '1991-04-02', 2985.00, NULL, 20),
-> (8654, 'MOMIN', 'SALESMAN', 8839, '1991-09-28', 1250.00, 1400.00, 30),
-> (8698, 'BINA', 'MANAGER', 8839, '1991-05-01', 2850.00, NULL, 30),
-> (8882, 'SHIVANSH', 'MANAGER', 8839, '1991-06-09', 2450.00, NULL, 10),

-> (8888, 'SCOTT', 'ANALYST', 8566, '1992-12-09', 3000.00, NULL, 20),
-> (8839, 'AMIR', 'PRESIDENT', NULL, '1991-11-18', 5000.00, NULL, 10),
-> (8844, 'KULDEEP', 'SALESMAN', 8698, '1991-09-08', 1500.00, 0.00, 30);

Query OK, 10 rows affected (0.008 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

1. Write a query to display EName and Sal of employees whose salary are greater than or equal to 2200?

```
SELECT EName, Sal FROM Empl WHERE Sal >= 2200;
```

OUTPUT:

```
MariaDB [LOANS]> SELECT ename, sal FROM Empl WHERE sal >= 2200;
+-----+-----+
| ename   | sal     |
+-----+-----+
| MAHADEVAN | 2985.00 |
| BINA      | 2850.00 |
| AMIR      | 5000.00 |
| SHIVANSH  | 2450.00 |
| SCOTT     | 3000.00 |
+-----+-----+
5 rows in set (0.001 sec)
```

2. Write a query to display details of employs who are not getting commission?

```
SELECT * FROM Empl WHERE Comm IS NULL;
```

OUTPUT:


```
MariaDB [LOANS]> SELECT * FROM Empl WHERE comm IS NULL;
```

empno	ename	job	mgr	hiredate	sal	comm	deptno
8369	SMITH	CLERK	8902	1990-12-18	800.00	NULL	20
8566	MAHADEVAN	MANAGER	8839	1991-04-02	2985.00	NULL	20
8698	BINA	MANAGER	8839	1991-05-01	2850.00	NULL	30
8839	AMIR	PRESIDENT	NULL	1991-11-18	5000.00	NULL	10
8882	SHIVANSH	MANAGER	8839	1991-06-09	2450.00	NULL	10
8888	SCOTT	ANALYST	8566	1992-12-09	3000.00	NULL	20

6 rows in set (0.001 sec)

3. Write a query to display employee name and salary of those employees who don't have their salary in range of 2500 to 4000?

```
SELECT EName, Sal FROM Empl WHERE Sal NOT BETWEEN 2500 AND 4000;
```

OUTPUT:

```
MariaDB [LOANS]> SELECT ename, sal FROM Empl WHERE sal NOT BETWEEN 2500 AND 4000;
```

ename	sal
SMITH	800.00
ANYA	1600.00
SETH	1250.00
MOMIN	1250.00
AMIR	5000.00
KULDEEP	1500.00
SHIVANSH	2450.00

7 rows in set (0.001 sec)

4. Write a query to display the name, job title and salary of employees who don't have manager?

```
SELECT EName, Job, Sal FROM Empl WHERE Mgr IS NULL;
```

OUTPUT:

```
MariaDB [LOANS]> SELECT ename, job, sal FROM Empl WHERE mgr IS NULL;
```

ename	job	sal
AMIR	PRESIDENT	5000.00

1 row in set (0.001 sec)

5. Write a query to display the name of employee whose name contains "A" as third alphabet?

```
SELECT EName FROM Empl WHERE EName LIKE '__A%';
```

OUTPUT:

```
MariaDB [LOANS]> SELECT ename FROM Empl WHERE ename LIKE '__A%';
Empty set (0.001 sec)
```

6. Write a query to display the name of employee whose name contains “T” as last alphabet?

```
SELECT EName FROM Empl WHERE EName LIKE '%T';
```

OUTPUT:

```
MariaDB [LOANS]> SELECT ename FROM Empl WHERE ename LIKE '%T';
+-----+
| ename |
+-----+
| SCOTT |
+-----+
1 row in set (0.001 sec)
```

7. Write a query to display the name of employee whose name contains “M” as First and “L” as third alphabet?

```
SELECT EName FROM Empl WHERE EName LIKE 'M_L%';
```

OUTPUT:

```
MariaDB [LOANS]> SELECT ename FROM Empl WHERE ename LIKE 'M_L%';
Empty set (0.001 sec)
```

8. Write a query to display details of employees with the text “Not given”, if commission is null?

```
SELECT EName, Job, Sal, COALESCE(Comm, 'Not given') AS Comm FROM Empl;
```

OUTPUT:

```
MariaDB [LOANS]> SELECT empno, ename, job, mgr, hiredate, sal,
-> COALESCE(comm, 'Not given') AS comm, deptno
-> FROM Empl;
```

empno	ename	job	mgr	hiredate	sal	comm	deptno
8369	SMITH	CLERK	8902	1990-12-18	800.00	Not given	20
8499	ANYA	SALESMAN	8698	1991-02-20	1600.00	300.00	30
8521	SETH	SALESMAN	8698	1991-02-22	1250.00	500.00	20
8566	MAHADEVAN	MANAGER	8839	1991-04-02	2985.00	Not given	20
8654	MOMIN	SALESMAN	8839	1991-09-28	1250.00	1400.00	30
8698	BINA	MANAGER	8839	1991-05-01	2850.00	Not given	30
8839	AMIR	PRESIDENT	NULL	1991-11-18	5000.00	Not given	10
8844	KULDEEP	SALESMAN	8698	1991-09-08	1500.00	0.00	30
8882	SHIVANSH	MANAGER	8839	1991-06-09	2450.00	Not given	10
8888	SCOTT	ANALYST	8566	1992-12-09	3000.00	Not given	20

```
10 rows in set (0.001 sec)
```